Lab 1 - Familiarising machine learning project

Exercise 1

In this exercise, you will do

- Create a IPython notebook in Google Colab, name it "CMT307Lab1Ex1.ipynb".
- Add a Text cell, write some text to describe the purpose of this notebook.
- Practice the following essential steps of a machine learning project.

```
"""
## Task 1: a very brief machine learning modelling process

*   Get data: the problem
*   Know the data: EDA
*   Process the data: pre-processing
*   Build model(s): modelling
*   Evaluate the model: evaluation
"""
from sklearn import datasets, svm
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np

pd.set_option('display.max_columns', None, 'max_colwidth', None, 'display.expand_frame_repr', False) # print all columns in full, prevent line break

"""## Get data"""

# get iris data
iris = datasets.load_iris()

# show the data
print(iris.data)
print(iris.target)

# split the data into input and output
X  = iris.data    # input
y  = iris.target  # output

n  = X.shape[0]   # number of data points
print(n)
```

```
# split the data set into training and test data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

"""## EDA"""

print(iris.DESCR)

# show two dimensions of the data, petal length and petal width

#plt.scatter(X_train[:, 2], X_train[:,3], c=y_train)   # petal length, petal width

plt.figure(figsize=(6, 4))

plt.plot(X_train[y_train==0, 2], X_train[y_train==0, 3], "bs")
plt.plot(X_train[y_train==1, 2], X_train[y_train==1, 3], "g^")
plt.plot(X_train[y_train==2, 2], X_train[y_train==2, 3], "y*")

plt.xlabel("Petal length", fontsize=12)
plt.ylabel("Petal width", fontsize=12)

plt.legend(['setosa', 'versicolor', 'virginica'])

"""## Data Preprocessing"""

from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()
scaler.fit(X_train)

X_train_scaled = scaler.transform(X_train)
X_test_scaled = scaler.transform(X_test)

"""## Data modelling"""

# specify a classification model of support vector machine, using default hyperparameters
#clf = svm.LinearSVC()
clf = svm.SVC()

# train the classifer on the training data
clf.fit(X_train_scaled, y_train)

# In this first example, the classifier is fitted using default hyperparameters.
# Note hyperparameters are usually needed to be optimised.
```

```
"""## Model evaluation"""

# use the trained model to predict iris types of test data
y_pred = clf.predict(X_test_scaled)
y_pred

from sklearn.metrics import accuracy_score

# calculate prediction accuracy
accuracy_score(y_test, y_pred)
```

Exercise 2

In this exercise, you will download from Learning Central the following files including one IPython notebook file and one data file to your local machine:

IPython Notebook:

CMT307Lab1Ex2-1.ipynb

Data files:

online_shoppers_intention.csv

You then upload the data to your Google drive and Colab, and study code cells of the notebook, make changes if necessary.

Note: The data set is from UCI Machine Learning Repository https://archive.ics.uci.edu/ml/datasets/Online+Shoppers+Purchasing+Intention+Dataset#, visit the web page for information about the data.